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Changing recess geographies: children's perceptions of a schoolyard renovation project promoting physical activity

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ABSTRACT

Insights into children's perceptions of their changing recess geographies are absent. The current paper provides qualitative evidence from 5 renewed schoolyards by investigating children's perceptions of their new schoolyards. Ten focus groups (2 at each school) including go-along interviews were conducted, in total 57 4 to 8 graders (28 girls). The functional aspects of the altered child–environment relationship were assessed using Gibson's concept of affordance, and changes in the meanings ascribed to the schoolyards by the children were studied by engaging a concept of place. Our findings reveal that schoolyards offering multiple potential affordances for physical activity seem more likely to engage a broader group of children in recess physical activity. The analysis, however, could point to a difference between places designed for children's play and places where children actually play. To optimise actualisation of the potential affordances for recess physical activity, involving children throughout intervention planning and decision-making is crucial.

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Introduction

Physical activity (PA) is crucial for children's health status (Andersen et al. 2004; Janssen and Leblanc 2010), but a large number of schoolchildren do not reach the recommended minimum level of 60 minutes of moderate-to-vigorous physical activity (MVPA) per day in Denmark and other Western countries (Trembley et al. 2016).

The schoolyard is an expedient setting for the promotion of children's PA (Ridgers et al. 2007; Cook et al. 2013; Martinez-Andres et al. 2012). Schoolyards are important childhood landscapes in terms of the considerable time spent there (Broekhuizen, Scholten, and de Vries 2014), the potential to reach and influence a large number of children from all backgrounds (Dobbins et al. 2013; Ridgers, Stratton, and Fairclough 2006), and for many children, schoolyards are the primary place they play outside (Tranter and Malone 2004). However, gender differences in PA behaviour in schoolyards are found to be extensive in schoolyards during recess. Many studies ascribe the gender difference in recess PA to the types of activities boys and girls engage in (Nielsen, Pfister, and Andersen 2011; Blatchford, Baines, and Pellegrini 2003; Boyle, Marshall, and Robeson 2003). But also, the design of the schoolyard often appealing mostly to boys' play has an important part to play (Harten, Olds, and Dollman 2008; Andersen, Klinker, et al. 2015; Nielsen, Pfister, and Andersen 2011; Karsten 2003).

A growing body of evidence identifies numerous ways in which renewed schoolyards can affect recess PA (Brink et al. 2010; Colabianchi et al. 2009; Anthamatten et al. 2011; Ridgers et al. 2007; Stratton and Mullan 2005). Studies have found that both boys and girls can exhibit greater schoolyard utilisation in renewed schoolyards (Anthamatten et al. 2011) but that boys are more likely than girls to exhibit greater schoolyard utilisation in renovated schoolyard environments (Anthamatten et al. 2011; Colabianchi et al. 2009). Predominantly, previous studies have employed objective measurements (e.g. heart rate and accelerometer) to evaluate the effect of schoolyard interventions (Parrish et al. 2013), whereas qualitative inquiries into this field are remarkable scarce (Hyndman 2015; Parrish et al. 2013; Ridgers, Salmon, et al. 2012). Consequently, only a unilateral evidence base to guide future schoolyard interventions is available, and, in particular, insights into children's perceptions of their changing recess geographies are absent. To address this gap, the current paper investigated children's perceptions of their new schoolyards, providing qualitative evidence from a post-intervention evaluation of a schoolyard renewal project, The Activating Schoolyards Study. Differences in gender were explicitly explored.

Theoretical framework

The paper seeks new and fertile ground within the field by employing two theoretical perspectives that highlight, respectively, the functionality and the meaning of the spatial changes in the schoolyards. While the functional aspects of the altered child–environment relationship were assessed using Gibson's (1979) concept of affordance, the changes in the meanings ascribed to the schoolyards by the children were studied by engaging a concept of place.

Affordance

The physical environment caters for human behaviour in various ways. Children experience opportunities and challenges according to the qualities of the setting, and the functional applications of the environment are critical in comprehending the child–environment relationship (Fjørtoft 2004; Heft 2010). In the current study we were inspired by Gibson's Affordance Theory positing that the physical environment affords different actions and behaviours; or in other words different environments 'invite' to various actions and activities (Gibson 2015, 1979). Affordances are relational, as they encompass not only features of the environment but also of the individual and is located at the interface between the environment and the person (Gibson 1979; Heft 1988). Accordingly, affordances offered by the environment are relational to the physical dimensions and abilities of the person in the environment (Heft 1988), meaning that they occur as a relationship between a feature in the environment and the intensions, perceptions and abilities of the person visiting the environment. Inspired by Heft (1988), Kyttä (2004) further elaborated this theory by distinguishing between potential affordances; infinite opportunities inherent in the qualities of the environment, and actualised affordances; opportunities for actions and behaviours in the environment that are perceived and recognised by the users. The actualised affordances can be considered active if they are used or passive if they are perceived but not used (Kyttä 2004).

The study ties in with the concept of affordance in two ways. First, informed by the theory of affordance it was hypothesised that introducing more variety of play environments in the schoolyard would increase the potential affordances for PA in the schoolyard. And further, by employing a participatory approach to the schoolyard renewal process, these new affordances being actualised by the children seemed highly plausible. Second, the concept of affordances is used to highlight and discuss how the children utilised the functional potentials of the new play environments and features in the renewed schoolyards. PA behaviours do, however, not result solely from a functional relationship between the children and their immediate environments. Indeed, individual, social and cultural factors conspire to regulate which affordances children can perceive and actualise (Kyttä 2004), and

further everyday environments, as schoolyards, are ascribed meaning and become places in children's lives.

Children's places

The essential role places play in children's geographies is well established (Hart 1979; Moore 1986; James 1990; Sobel 1993; Nabhan and Trimble 1995). Drawing on phenomenological ideas, the concept of place sheds light on intimate, personal and emotional relationships between person and place (Gregory et al. 2009; Cresswell 2015). Further, there is an increasing interest in children's use of place for play in outdoor environments (Moore 2015; Ridgers, Knowles, and Sayers 2012; Holt et al. 2015; Veitch, Salmon, and Ball 2007), particularly in children's experiences within adult-constructed places for children's play (Moore 2015; Wake 2008). Numerous researchers have highlighted the value children attribute to informal places for play created of their own (Rasmussen 2004; Arvidsen 2018; Moore 2015). However, one central issue to this is the marked decrease in opportunities for children to construct their own places for play because of powerful adult discourses of regulation and protection (Frost 2007). According to Rasmussen (2004), children's places are often unnoticed by adults, and the significance of these places is in danger of being ignored because only children fully appreciate the value of these places. Furthermore, children's places are in danger of being tidied up when they exist in spaces which are managed by adult rationales (Rasmussen 2004; Moore 2015). In this paper, the notion of place is employed to consider children's emotional relations to the renewed schoolyards while paying particular attention to unintentional violations of children's places arising from the schoolyard renewals.

Methods

This study, drawing on a qualitative post-intervention dataset, is part of a quasi-experimental schoolyard intervention study, the Activating Schoolyards Study, investigating the impact of renewed schoolyards on children's (grades 4–8; 10–15 years old) recess PA behaviour. PA was measured objectively using accelerometers and GPS at baseline and follow-up (Andersen, Pawlowski, et al. 2015). Before and after intervention we also subjectively explored the factors children perceived as influencing their recess PA by interviewing the children (Pawlowski et al. 2016). The current study draws on our qualitative post-intervention dataset.

Schools included

The schools were selected based on an open call for proposals issued by a partnership consisting of three non-governmental organisations (The Danish Cancer Society, The Danish Foundation for Culture and Sport Facilities, and the Danish foundation Realdania). In October 2012, all public primary and lower secondary schools in Denmark (approximately 1800 schools) were invited to submit a vision proposal. Among 106 submitted proposals the partnership selected seven projects for realisation in February 2014. One school was delayed beyond the study timeline and one school declined to participate in the current study.

The 5 schools included in the current study varied in geographic location, school type (3 rural and 2 urban schools), number of pupils enrolled (93–670 children), socioeconomic status (3 schools higher and 2 schools lower than average income rate in Denmark), and size of the schoolyard (6747–59,333 m²). Approximately 60 minutes per day were allocated to recess, distributed over two to four periods, including the lunch break, lasting 25–30 minutes. Recess was typically characterised by free play without any organised curriculum or teacher involvement, aside from supervising the children and handling conflicts and accidents. Three of the schools had a policy making outdoor recess policy obligatory for children up to grade seven.

The renewal process

The renewal took place during 2015. The schoolyards were renewed using a participatory bottom-up approach involving children in the development process. At some schools, architects were responsible for workshops with children and at other schools a process consultant was hired to facilitate the workshops. Number, age and characteristics of children included in the process differed among the schools so did the time spent, and co-design techniques used to develop with the children. At all schools, the children were only part of the first step of the development process creating ideas that were handed over to the architects designing the schoolyards. It was up to the schools to decide how the children participated in the process reflecting local challenges and needs, and therefore both process and outcome were highly tailored. At some of the schools, the interventions took place in the existing schoolyard, whereas other schools expanded their outdoor area by including adjacent spaces (e.g. woodland or parking area). Even though the design and dimension of the intervention components varied widely, some features were present in several schools, e.g. climbing walls, balance-bars, theatre/dancing-stages, skating areas, trampolines, hills, and ball game features. The total budget for each of the 5 schools ranged from 100,000 to 530,000 EUR. The partnership funded 50% of the budgeted cost of the renovations and the schools provided the remaining 50%.

Data collection and participants

Data were collected between April and June 2016. The renewed schoolyards primarily targeted children in grades 4 through 6 (age 10–13) since PA declines significantly during these school years (Nader et al. 2008; Dumith et al. 2011). At 4 schools, data were collected among grades 4 through 6 children. At one school the data collection included children in grades 7 and 8 (age 13–15) because the renewed schoolyard at this particular school had this target group.

To obtain an in-depth insight into children's perceptions of their renewed schoolyards, conducting focus groups with children was selected as the most suitable method for this study (Horner 2000; Krueger and Casey 2002). Focus groups were conducted by a trained group interviewer (the first author) during school hours at all schools and lasted for approximately 45 minutes. Each focus group intended to include two same-gendered children from each grade level randomly sampled from returned consent forms. In total 10 focus groups (1 boy and 1 girl focus group at each school) were conducted including 57 children (28 girls). All children were of Danish origin.

As part of the focus groups, the children took the interviewer on a walking tour around their schoolyard, similar to a go-along interview approach (Carpiano 2009; Kusenbach 2003). The children were asked to guide the interviewer around the schoolyard and decide the route themselves. Through the guided walking tour, the interviewer asked questions prompted by what she observed, not systematically following an interview guide. Questions during the walk included for example, 'Are you here during recess?' 'What are you doing here?' 'Why are you not here during recess?' 'What do you think about your schoolyard?' The main advantage of go-along interviewing was that the interviews could demonstrate the children's perceptions, experiences, interactions and practices within both renewed and non-renewed places in the schoolyard environment (Carpiano 2009; Kusenbach 2003). The focus groups were filmed using an iPad mini® to record both the verbal and non-verbal interaction of participants.

Data analysis

A thematic analysis of the focus group transcripts was conducted with the explicit purpose of identifying the children's perceptions of the renewed schoolyards (Neergaard et al. 2009; Sandelowski 2000). Phrases from transcripts that referred to the children's perceptions of the renewed schoolyard were highlighted, from which themes and subthemes emerged. All parts of the data analysis included

a gender perspective focusing on similarities and differences between boys' and girls' perceptions of the renewed schoolyards.

Ethics

Prior to the data collection, all school principals and involved teachers approved the study. We then prepared a consent form explaining all our data collection procedures related to the use of accelerometer, GPS and focus groups, respectively. We clearly mentioned our use of tablets to film the focus group interviews. If the parents allowed their child to participate in the focus groups, they ticked off a specific box on the consent form. We randomly selected our participants from the returned written informed consent forms from parents who had ticked off the specific box. All invited children provided their assent for participating. They were informed about the purpose of this particular study and were told that they could withdraw from the study at any time and that pseudonyms were used for anonymity.

The study and its data-management procedures were registered and approved by the Danish Data Protection Agency (2014-41-2801) and ISRCTN (ISRCTN17944407). According to the Danish National Committee on Health Research Ethics, obtaining formal ethical approval was neither possible nor required as the project was not a biomedical research project.

Insights from the study discussed

Gendered places

All children were generally enthusiastic about the renewal and appreciated the abundance of new and appealing play environments in the schoolyards. However, the renewals seemed mostly to affect girls' recess PA behaviour. The idea that places are gendered is not new to children's geographers (Thorne 1993; Holloway and Valentine 2000). In line with numerous studies (Harten, Olds, and Dollman 2008; Andersen, Klinker, et al. 2015; Nielsen, Pfister, and Andersen 2011; Spark, Porter, and de Kleyn 2018; Karsten 2003), girls described the pre-intervention schoolyards as a 'place for boys' to play soccer with no or very few play environments motivating girls to be active. After renewal, the girls experienced that the schoolyards to greater extent included play environments aimed at girls motivating them to be active in the schoolyard. A girl from grade 5 voiced it like this: 'There are more opportunities for girl play' (Jasmine, school 4). Furthermore, a seventh-grade girl from another school exemplified what she defined as a play environment for girls: 'Actually the spot with those mirrors and all that, where you can play music and dance is really nice and girlish' (Jessica, school 1). Additionally, to the dance spots, environments that featured trampolines, small treetop houses, and gymnastic mats were examples of play environments in the renewed schoolyards mentioned as appealing by girls. In contrast, most boys explained that they played soccer during recess both before and after renewal and therefore the presence of more variety in play environments had not changed these boys' PA behaviour during recess. A boy from grade 6 stated: 'It's [the new schoolyard] really good, but I don't use it' (Jacob, school 3).

Tranter and Malone (2004) state that different children prefer different types of play environments and it seems plausible that there are gender differences in how children utilise specific play environments in the schoolyard (Anthamatten et al. 2011; Nielsen, Pfister, and Andersen 2011). Other studies have found that boys are more likely than girls to exhibit greater schoolyard utilisation in general but also in renovated schoolyards with new play environments (Anthamatten et al. 2011; Colabianchi et al. 2009). However, like in our study, Anthamatten et al. (2011) also observed greater schoolyard utilisation among girls on renovated schoolyard sites offering a greater diversity of play opportunities than traditional schoolyards.

Following Gibson's (1979) notion of affordance, children perceive possibilities in what the environment offers and interpret and use the environment they encounter in a functional manner

based on their abilities and background, which can explain the gender differences found. In a previous study, we realised that children from an early age learn what it means to act as a boy or a girl, respectively, which influenced how boys and girls interpreted and used the environment they encountered in the schoolyards (Pawłowski et al. 2015). Girls' generally lower recess PA in the schoolyard (Ridgers, Salmon, et al. 2012) could be ascribed either to limitations in the functionality of the schoolyard environment or to confined perceptions and abilities among the girls inhabiting the schoolyard environment. Thus, to actualise affordance for girls' recess PA more different play environments appealing to different children should be incorporated in the schoolyard design. However, for a deeper understanding of girls' abilities and needs of a play environment listening to girls is crucial. Having said that, extending children's abilities to engage with a broader range of play environments and to participate in less gender segregated and stereotyped play activities may also affect recess PA behaviour.

Failing to see children's places

According to both genders, several of the renewed schoolyards contained environments with innate play potentials that were declared out of bounds. These restrictions reduced the children's freedom to explore their schoolyards opposing actualisation of the potential affordances for recess PA (Kytä 2004). Like in other studies (Ridgers, Knowles, and Sayers 2012; Staempfli 2009), the children were not being afforded the opportunities to play in natural environments. At some schools, children reported how new play environments introduced through the renewal process had spoiled places, which previously was much appreciated by the children. At school 2, a nature-based environment previously used for car parking, where the oldest children were allowed to hang out during recess, was turned into a multicourt for all children. A girl from grade 6 stated: 'I think it is annoying that they removed the parking spot. It was our place and now we miss a place only for us' (Maria, school 2). At school 5, the children missed a woodland area like it was before renewal. A conversation between the moderator and two boys from grades 5 and 6 at school 5 highlighted this:

- Jonathan: I really liked it before.
 Moderator: What did you like about it?
 Jonathan: We played war games all the time up in the woodland. We found sticks we used as machineguns and we lay hidden in the edge of the woodland and then there was a slope with a rope we claimed.
 Moderator: But can't you still play war games in the woodland area?
 Jonathan: It's not the same now with all that asphalt.
 Moderator: What was it that made it so nice to play there?
 Jonathan: Well, just that there were not so many things there, so you had to make-up things yourself and visualise things yourself. It was just a big area where you could run around doing everything imaginable.
 Charlie: Yes, I liked that you could make up your own games there.

Changing these natural environments into well-equipped play environments attracted many of the youngest schoolchildren (grades 1 through 3 children), which had negative implications for the older children for whom these places were used to cater silence, privacy, fantasy and self-invented games. Our findings resonate with Herrington and Brussoni (2015) who stated that the character and context of natural environments can offer other types of stimuli than the traditional schoolyard environment.

In line with what Rasmussen (2004) and Moore (2015) have highlighted, we found numerous examples of how new play environments and adult rationales worked against children's places in the renewed schoolyards. Also, other schoolyard studies contemporarily designed have identified vast incongruence between adults' conception of the optimal play environments and the needs of the children (Brown and Burger 1984; Christiansen et al. 2017). Similar to Moore (2015) many children resisted the powerful adult discourses of regulation by finding physical or symbolic privacy even when opportunities to do so were limited in space and affordance within the schoolyard. But these

alternative places were often indoors where the children used computers or smartphones to create their own place.

We also found play environments designed by adults to be preferred beyond places defined by the children themselves. For example, a girl from grade 5 explained: 'Now where we have got foursquare pitches we play foursquare games much more' (Sarah, school 4). The marked foursquare pitch seemed to actualise affordance for foursquare play easier than before renewal where the children had to define and mark the place themselves with chalk. Also, the new play environments were used to more creative play not directly associated with the given function of the feature, expressed like this by a boy from grade 4: 'We first played hide-and-seek when we had the new schoolyard because we got better and more hiding places. Before we had only asphalt and that was not very nice' (Daniel, school 3). Besides hide-and-seek, children also talked about new role-play and tag games developed in the renewed schoolyards.

Our examples above show that if affordances for recess PA in the schoolyard are to be actualised, we need to identify why children prefer and engage with a certain play environment in the schoolyard. It is important to try to understand children's places by being aware of not solely the physical structures of the environment, but also paying attention to how a certain play environment matters – how it shapes social interactions and how it is made (Tranter and Malone 2004).

Child-designed places

Both insights above emphasised the importance of listening to children when designing schoolyards. Over the last decades, it has been more common that children are actively engaged in planning and designing their physical environments (den Besten, Horton, and Kraftl 2008). Involving children has been found to develop a sense of ownership and a feeling of being taken seriously (Parnell, Cave, and Torrington 2008; Hart 1992). However, Cele and van der Burgt (2015) stated that many projects using a 'participatory' approach with children do not reflect a true willingness to understand what children are expressing. Evidence suggests participatory approaches may give rise to feelings of frustration rather than empowerment if it later transpires that most of the design decisions had been made without regard to children's suggestions and perspectives (Parnell, Cave, and Torrington 2008). Similar findings appeared in the current study, where particularly boys had expected more or something different in the schoolyard based on children's suggestions during the participatory design process. At school 3, two boys from grade 6 explained it like this:

- Peter It's a little disappointing. All the good advices we had come up with and then it's just a fourth of it that has been realised.
- Jacob Maybe the expectations have been too high.
- Peter Yes, and they didn't build all we were told. For example, there should have been a big box where we could sit and listen to music. It isn't there and it could have been cooler with two big hills instead of all the small slopes.

As illustrated above, the disappointment covered varied frustrations related to the participatory process. Some felt that they had not been listened to while others felt that they had not gotten what they had been promised. However, most of the dissatisfied children were disappointed with a specific play environment – often a specific ball game court or feature not having standard dimensions. For that reason, it was experienced not to function as it should in practice opposing actualisation of the potential affordances of the given play environment or feature. Two boys from grades 5 and 6 (School 2), respectively, explained what specifically disappointed them in their renewed schoolyard:

- Elliot The multicourt is a little too narrow. It could be nice if it was a little bigger.
- William Yes, and these open gates they are really annoying because the ball gets out all the time.

Cele and van der Burgt (2015), found it very important to take children's participation one step further and include them in decision-making. With that in mind, a critical point of our project was that the children only took part in the first step of the intervention development, creating ideas,

which were then handed over to the architects for further development and they were not part of the decision-making processes. Also, according to Gibson (1979), actualisation of affordances can involve a longer process of adjustment between children and their environment. Therefore, children need to be listened to throughout all project stages through methods and practices that suit them, to optimise actualisation of the potential affordances for PA. However, studies involving children in schoolyard design have found difficulties in optimal participation caused by lack of time, resources and guidance (den Besten, Horton, and Kraftl 2008; Parnell, Cave, and Torrington 2008). Therefore, future schoolyard renewal studies should investigate if more extensive dialogue with children led by trained facilitators throughout the developing and decision-making processes can help architects to design schoolyards that are perceived more positively by the children and hereby actualise the potential affordances for recess PA.

Strengths and limitations of our study

Focus groups are an effective method to gain insight into schoolchildren's perceptions and experiences of selected topics such as health (Albrecht, Johnson, and Walther 1993). Children age 10–15 years old are undergoing a developmental transition from child to adolescent, which means that they tend to be reticent when talking with adult strangers. However, when interviewing groups of peers, they are more relaxed and willing to share perceptions (Horner 2000). A limitation of using focus groups is that the results are not generalisable due to the fact that the data collection methods were not systematic (Fern 2001). However, our aim was to outline relevant key issues which might inform further research, and generalisability was not an expected attribute. This qualitative study should be seen as the supplement requested to the many quantitative evaluation studies in the field of schoolyard PA (Hyndman 2015). Another weakness of conducting focus groups with children is related to group composition. Status in the peer group can contribute to polarisation within the group if dominant children control the topics discussed (Morgan 1996). However, the go-along interview allowed for individual talks with the children to reduce this disadvantage. Using go-along interview in the focus groups allowed for triangulation of the results and added to the reliability of the study (Creswell 2009) and gave a more complete clarification of the field (Darbyshire, MacDougall, and Schiller 2005; Shenton 2004).

We wanted to gain insights from a user perspective, which means that we chose only to interview the children who the intervention was aimed at. Seen in retrospect, it would also have been fruitful to interview the younger schoolchildren because of their extensive use of the new play environments in the schoolyards. Moreover, including teachers' view could possibly have broadened our insights into the renewed schoolyards. However, according to 'the new child paradigm', it strengthened our study to acknowledge children as individuals, independent of the perspective of adults (James and Prout 2005). In addition, we experienced that the children had clear opinions on the renewed schoolyards and in line with other child studies, they were delighted for their voices to be heard (MacDougall, Schiller, and Darbyshire 2004; Darbyshire, MacDougall, and Schiller 2005; Morrow 1998).

Conclusion – and then what?

This paper reflected empirically and critically upon the Activating Schoolyards Study by investigating children's perceptions of their schoolyards after renewal. A post-intervention study on the children's perceptions of their renewed schoolyards is necessary to gain more comprehensive evaluations of schoolyard interventions that can aid the development of future PA interventions in the schoolyard environment. The types of play environments 10–15-year-old schoolchildren enjoy during recess differ based on abilities and perceptions and thus differs between genders. For that reason, the schoolyards that extended their variety in play environments engaged a broader group of children in PA. In particular, natural environments without playground features seem to be

important environments also to prioritise in the schoolyard to allow for other types of stimuli than the traditional schoolyard environment.

The analysis, however, could point to a built-in conflict between places for children's play and children's play places. We found that children were disappointed with new play environments in the schoolyard as they did not match their expectations, and the new schoolyards may possibly still contain innate unexploited potentials adults are unaware of and thus not supported by adults. We, as adults, are challenged to think differently when laying out places for children's play. Children's perceptions and their social structures add value in the planning and developing of environments provided for them, such as schoolyards. We need to insist on the importance of listening to them, even (perhaps especially) when their views challenge conventional adult wisdom. It requires us to reify children as independent knowing persons whose voices add something important to debate. Therefore, future schoolyard renewal studies promoting recess PA should investigate if extensive dialogue with children led by trained facilitators throughout the developing and decision-making processes can help architects to design schoolyards more positively perceived by the children and hereby actualise the potential affordances for recess PA.

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Disclosure statement

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